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| <div> <div>AJAY VENKATESH</div> <div> +1 516 585 9013 Email LinkedIn GitHub Portfolio </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                               |                                      |
| Education                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                               |                                      |
| New York University   <i>Master of Science in Computer Engineering (GPA – 3.9/4)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                               | Sep 2024 – May 2026                  |
| • <i>Coursework:</i> System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                               |                                      |
| Skills                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                               |                                      |
| <b>Programming Languages:</b> Python, Java, C++, TypeScript, JavaScript (ES6), SQL<br><b>Web Technologies:</b> REST API, GraphQL API, Microservices, React, Node.js, Express.js, Sequelize, Django, FastAPI<br><b>Database:</b> RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse<br><b>Cloud:</b> AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs), Kubernetes<br><b>Tools:</b> Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Power Automate, Power BI, QuickSight<br><b>AI/ML:</b> PyTorch, Hugging Face, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, SVM, Random Forest, MLflow, PySpark                                                                                                                                                                                                                                                                                                                                                              |                               |                                      |
| Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                               |                                      |
| CampusMesh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                               | Jan 2026 – Present                   |
| <b>Software Engineer</b>   <i>Real-time microservices, recommendation engine, and LLM-based AI retrieval pipelines</i><br>– Developed scalable <b>Node.js microservices</b> powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by <b>MySQL (Sequelize)</b> and <b>DynamoDB</b> .<br>– Deployed <b>serverless architecture</b> on AWS using <b>Serverless Framework</b> with <b>Lambda, API Gateway, S3, and CloudFront</b> across <b>4 environments</b> (local, dev, staging, production); enforced security with <b>KMS-managed keys, SSM Parameter Store</b> , and <b>least-privilege IAM</b> .<br>– Worked on <b>LLM-based retrieval pipelines (RAG)</b> for CampusIQ (AI Assistant), leveraging <b>embedding-based semantic search</b> and structured data indexing to improve contextual accuracy by <b>30%</b> .                                                                                                                                                                                                            |                               |                                      |
| Amazon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                               | May 2025 – Aug 2025                  |
| <b>Software Development Engineer Intern</b>   <i>Serverless AWS infrastructure, React 18 frontend, and Java backend systems</i><br>– Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the <b>system design</b> for approval and established Role Based Access ( <b>4+ roles</b> ) using <b>OAuth, Midway SSO, and HELIS</b> .<br>– Provisioned <b>serverless Infrastructure-as-Code</b> using <b>AWS CDK</b> with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by <b>70%</b> and enabling <b>multi-stage, multi-region CI/CD deployments</b> .<br>– Built a <b>React 18</b> frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by <b>40%</b> .<br>– Engineered backend services in <b>Java</b> using <b>Coral, Smithy, Dagger</b> , and <b>CBOR serialization</b> ; <b>applied low-level design and design patterns</b> reduced backend execution time by <b>35%</b> and ensure reliability by unit and integration testing using <b>Mockito</b> . |                               |                                      |
| New York University (Novel AI Technologies, Inc.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                               | Aug 2024 – Present                   |
| <b>Software Developer (On-Campus)</b>   <i>NSF-funded AI health platform, ETL pipelines, and full-stack insurance analytics</i><br>– Architected an <b>NSF-funded (\$100K)</b> AI-powered health app for <b>iOS</b> and <b>Android</b> , processing real-time wearable data; deployed cloud-hosted AI models for health matrix analysis and process food images.<br>– Engineered ETL pipelines transforming <b>NoSQL</b> into <b>PostgreSQL</b> using <b>concurrency control and locking mechanisms</b> ; processed <b>250K+</b> records and powered <b>AWS QuickSight</b> dashboards for real-time health analytics.<br>– Developed an insurance analytics platform ( <b>React, Node.js, REST APIs</b> ); deployed ( <b>Docker</b> containers on <b>EC2</b> , implemented Stripe billing, RBAC, and row-level security, generating <b>\$50K+</b> revenue.<br>– Streamlined underwriting via broker portal integrated with insurer platform; implemented <b>document and media processing</b> and <b>WebSocket-based real-time communication</b> ; secured <b>6 insurance clients</b> .                   |                               |                                      |
| PricewaterhouseCoopers (PwC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                               | Aug 2021 – Aug 2024                  |
| <b>Senior Software Engineer</b>   <i>Cloud-native backend, NLP/OCR automation, and data engineering on Azure</i><br>– Developed a distributed backend on <b>Microsoft Azure</b> for a health platform, integrating with <b>Microsoft CRM</b> and implementing event-driven processing and fault tolerance.<br>– Led code reviews and mentored engineers on <b>software design principles</b> , reducing production bugs by <b>25%</b> .<br>– Deployed an <b>NLP based OCR neural model</b> on Azure to automate structured data extraction, reducing data processing time by <b>75%</b> .<br>– Optimized data ingestion with <b>parallel processing</b> for <b>100K-row</b> from Excel; reduced execution time from <b>5 hours to 1 hour</b> .<br>– Built <b>Power BI dashboards</b> processing large datasets from multiple databases, improving reporting efficiency by <b>40%</b> .                                                                                                                                                                                                                    |                               |                                      |
| Projects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                               |                                      |
| <b>Personal Finance Analyzer (Python Data Analysis Project)</b>   <i>Python, Pandas, NumPy, Matplotlib, Seaborn</i><br>– Built a data-driven application to analyze personal financial transactions and categorize expenses using <b>Pandas</b> for efficient data processing.<br>– Generated interactive visualizations and spending insights using <b>Matplotlib</b> and <b>Seaborn</b> to identify trends and optimize budgeting.<br>– Designed reusable and modular code structure following <b>OOP principles</b> to improve maintainability and scalability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                               |                                      |
| Certifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                               |                                      |
| AWS Cloud Practitioner                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Azure Data Fundamentals       | Microsoft Solution Architect Expert  |
| Power BI Data Analyst Associate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Microsoft Developer Associate | Stanford Machine Learning (Coursera) |